Digital Systems

Lecture 15 & 16

Scheduling Processes & Messages

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Following the course of past years by Mike Spivey

C Pointer Syntax

- Variable containing 1

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- Pointer to `distance`-typed variable, assign address of `a`

  `distance* b; b = &a;`
- Content of address in b

  `(*b).inch == a.inch;  /* Or... */ b->inch == a.inch;`

Today

How are processes started?

What happens to the Operating System on startup?

Starting a Process

The first time a process runs, it is resumed just as if returning from a system call.

So we set up a fake exception frame that invokes the process body when resumed.

- `r0` = integer argument,
- `pc` = process body,
- `lr` = address of exit stub, in case body returns.

Creating “fake” Stack Frame

```
int start(...) {
    ...
    unsigned *sp = p->sp - FRAME_WORDS;
    memset(sp, 0, 4*FRAME_WORDS);
    // Macros for offsets in stack, ***_SAVE
    sp[PSR_SAVE] = INIT_PSR;  // Thumb bit switched on
    sp[PC_SAVE] = (unsigned) body & ~0x1; // Activate the proc
    sp[LR_SAVE] = (unsigned) exit; // Make it return to exit()
    sp[R0_SAVE] = (unsigned) arg;  // Pass arg in r0
    sp[ERV_SAVE] = MAGIC;
    p->sp = sp;
    ... }
```

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To start, just initiate context switch by returning from `system_call()`.

- Q: Return value?

Starting the OS

After `init()` called, all tasks have stack frames that will allow a context switch into them.

```
void __start(void) {
    /* Create idle process */
    ...
    /* Call the application's setup function */
    init();
    /* The main program morphs into the idle process. */
    os_current = idle_proc;
    set_stack(os_current->sp);
    idle_task();
}
```

Idle Process

```
void idle_task(void) {
  /* Pick a genuine process to run */
  yield();

  /* When there's nothing to do: */
  while (1) pause();
}
```

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- During context switch, interrupt stores some state on stack,
  assembly code stores the rest.

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We have already seen that we need to start processes, and keep track of various process-specific quantities (e.g. stack location), in order to do context switches. How does the OS do this?

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Process Structure

```
struct _proc {
    int pid;            // Process ID
    char name[16];      // Name for debugging
    unsigned state;     // SENDING, RECEIVING, etc.
    unsigned *sp;       // Saved stack pointer
    int priority;       // Priority: 0 is highest
    proc waiting;       // Processes waiting to send
    int pending;        // Whether interrupt pending
    int msgtype;        // Message to send or receive
    message *msgbuf;    // Pointer to message buffer
    proc next;          // Next process in ready/send queue
};
```

Possible states

- `DEAD`: process that has exited. It is also the value present in slots of the table that have never been used.
- `ACTIVE`: process is ready to run. The process is either the current process, or the ready queue waiting to become current.
- `SENDING`: process is waiting to send to another; the process will appear in the sending queue of the destination.
- `RECEIVING`: process is waiting to receive a message; the `msgtype` field will specify who may send to it.
- `SENDREC`: call to `sendrec`, combining the functions of send and receive.
- `IDLING`: idle process, which runs when no other process is ready.

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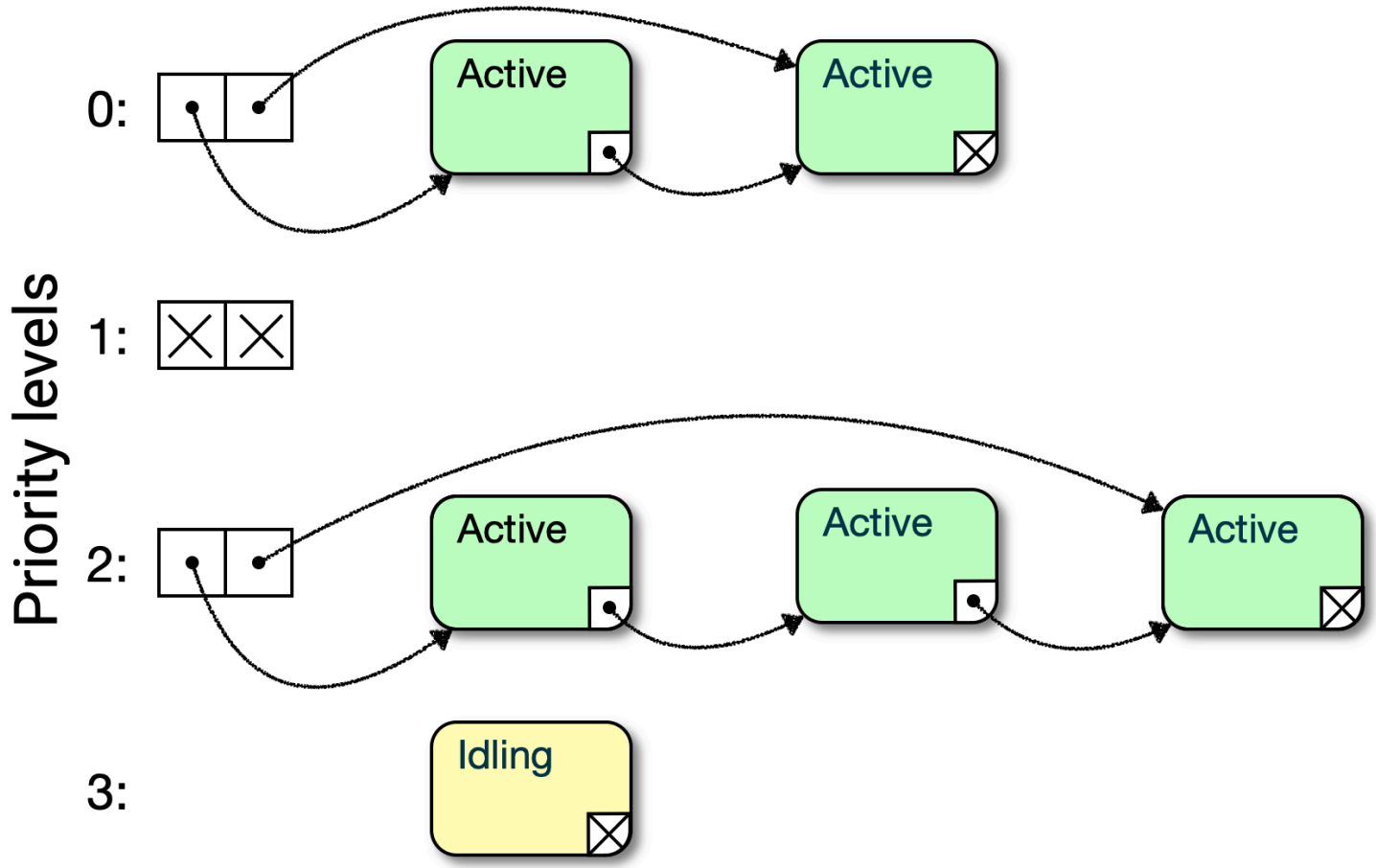
Draw memory layout (heap, stacks, ..., processes, main stack). ⟸

? How does the OS decide which process to run next?

Start with the easy case: context switch initiated by `yield()`.

Ready Queues

In `ACTIVE` processes, the ready queue is implemented using the `next` field.



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#define NPRIO 3       /* Number of non-idle priorities */
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    proc head, tail;
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```

Last line of `start(...)`, after setting up fake exception frame...

```
    make_ready(p);
```

Adds process to appropriate queue.

Adding to the Queue: `make_ready()`

```
static inline void make_ready(proc p) {
    int prio = p->priority;
    if (prio == P_IDLE) return;

    p->state = ACTIVE;
    p->next = NULL;    // Because last in queue
    queue q = &os_readyq[prio];
    if (q->head == NULL)
        q->head = p;
    else
        q->tail->next = p;
    q->tail = p;                          }
```

Full Story of Startup

Have code side-by-side.

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- Make current function the idle task.

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- Calls `make_ready()` to add process to appropriate queue.
  At initialisation, all processes are `ACTIVE` and in the ready queue.
- Make current function the idle task.
- `yield()`, so OS starts scheduling procedure.

Selecting Next Process

We end up in `system_call(...)` again.

```
unsigned *system_call(unsigned *psp) {
    short *pc = (short *) psp[PC_SAVE]; /* Program counter */
    int op = pc[-1] & 0xff;  // Syscall number from SVC instr
    os_current->sp = psp;    // Save sp of the current process
    ...
    switch (op) {
    case SYS_YIELD:
        // Add current to back of Q. Choose next process.
        make_ready(os_current); choose_proc(); break;
        ... }
    return os_current->sp; }  // Return sp for next process
```

Selecting Next Process

Find highest priority process to run, remove it from queue, `return`.

```
static inline void choose_proc(void) {
    for (int p = 0; p < NPRIO; p++) {
        queue q = &os_readyq[p];
        if (q->head != NULL) {
            os_current = q->head;
            q->head = os_current->next;
            return;
        }
    }
    os_current = idle_proc;
}
```

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void NOINLINE send(int dest, message *msg) {
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}
```

`hardware.h`:

```
#define syscall(op)     asm volatile ("svc %0" : : "i"(op))
```

System Call after `send()`

Just like `yield()`, `send()` simply causes a syscall interrupt.

```
unsigned *system_call(unsigned *psp) {
  short *pc = (short *) psp[PC_SAVE];
  int op = pc[-1] & 0xff;
  os_current->sp = psp;
  switch (op) {
  case SYS_SEND:
    mini_send(arg(0, int), arg(1, int), arg(2, message *));
    break; ...
  }
  return os_current->sp;
}
```

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Remember the rules for communication by messages!

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- Transfer message to destination. Destination `READY`. Add to queue.

The destination process cannot receive…

- Multiple possible reasons for this (Q: which?)
- Must (somehow) keep sender in queue until process is ready to receive it.

Sending Queues

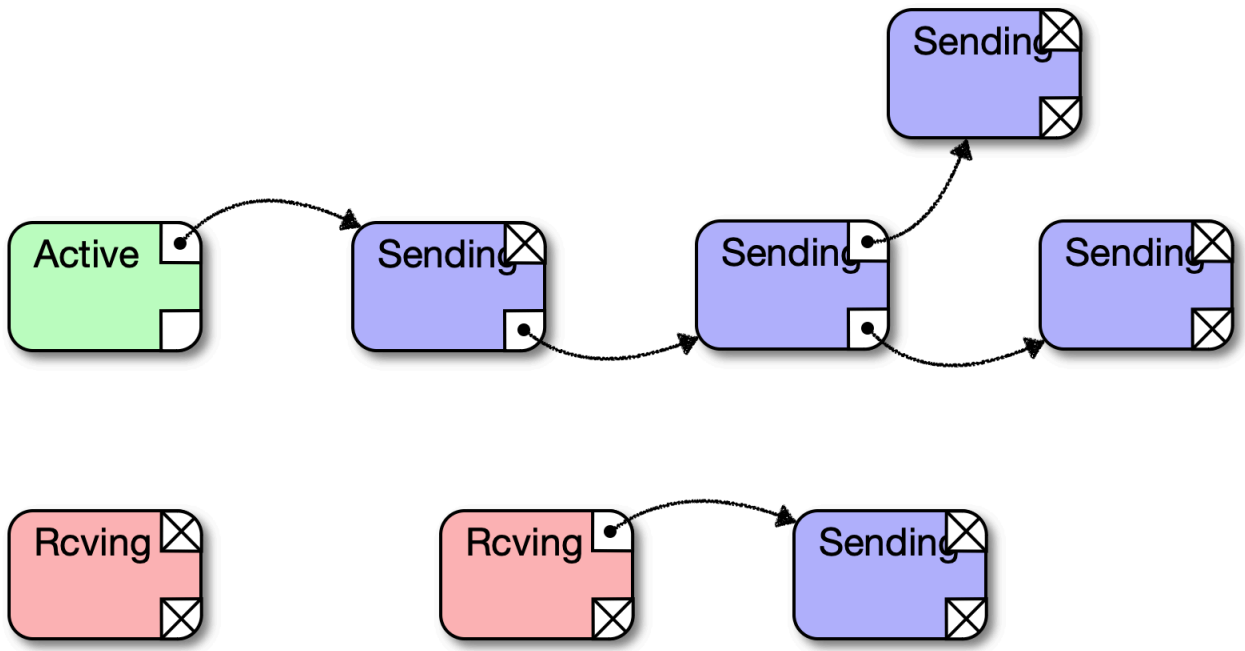
Processes can have multiple other processes waiting to send.

Next process trying to send is stored in `waiting`.

Rest of the send queue is chained through `next`.

Q: Does this conflict with the ready queue?

Each process has a queue of others waiting to send to it.



- Active process has 3 processes waiting to send to it.
- One process waiting to send also has a process waiting to send to *it*.
- One waiting to receive, but nothing to send to it yet.
- One waiting to receive, with one waiting to send to it, but not the right type of message.

System call after send()

```
static void mini_send(int dest, int type, message *msg) {
  int src = os_current->pid;
  proc pdest = os_ptable[dest];
  if (accept(pdest, type)) { // Receiver is waiting
    deliver(pdest->msgbuf, src, msg);
    make_ready(pdest); make_ready(os_current);  // *DISCUSS*
  } else {
    os_current->state = SENDING;
    queue_send(pdest);  // Sender waits. Join receiver's queue
  }
  choose_proc();
}
```

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Receiving Messages

```
static void mini_receive(int type, message *msg) {
  if (/* interrupt is due */) { // later...
  } else {
    proc psrc = find_sender(os_current, type);// search send Q
    if (psrc != NULL) { // Is a sender waiting?
      deliver(msg, psrc->pid, psrc->msgbuf);
      make_ready(os_current); make_ready(psrc);
    } else { // No luck: we must wait
      set_state(os_current, RECEIVING, type, msg);
  } }
  choose_proc();
}
```

? How are messages transferred between processes?

And in particular, how is this done in interrupts?

Interrupt Handler

```
void default_handler(void) {
    int irq = active_irq(), task;
    if (irq < 0 || (task = os_handler[irq]) == 0)
        panic("Unexpected interrupt %d", irq);
    disable_irq(irq);
    interrupt(task);
}
```

Discuss code.

How are messages delivered

How are messages delivered

```
static inline void deliver(proc pdest, proc psrc)
{
    if (pdest->msgbuf) {
        *(pdest->msgbuf) = *(psrc->msgbuf);
        pdest->msgbuf->sender = psrc->pid;
    }
    make_ready(pdest);
}
```

Summary

- Process table
- Process states
- Ready queues
- Send queues
- Implementation of delivery schedule of messages

Notes contain another example of a device driver, and a discussion of `SENDREC`. Do take a look at this.

End of Term!

Next term, start from the complete bottom of the stack:

? How to we build a computer?

... starting from a transistor